Bayesian Population Structure in Bacteria

DEGAiN training

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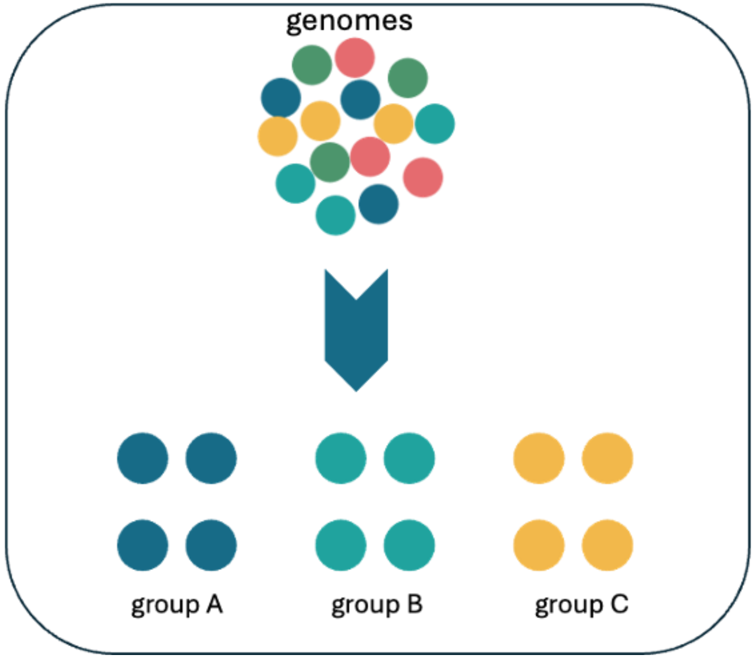
Learning objectives

- Define bacterial population structure and its importance.
- Know the principles of Bayesian population structure analysis.
- Differentiate between the pangenome, core genome, and accessory genome.
- Carry out genome annotation, pangenome analysis and core genome alignment
- Apply the workflow to infer and interpret bacterial population structure from genomic data.

What is Bacterial Population Structure?

The organization of bacterial genomes within a specie into a genetically related groups based on shared genetic variation and evolutionary history.



e.g. 100 E.coli isolates

Identical?

Genetic variation

Differences from?

- Mutation
- Recombination
- Horizontal gene transfer

Sheppard SK, Guttman DS, Fitzgerald JR. Population genomics of bacterial host adaptation. Nat Rev Genet. 2018 Sep;19(9):549-565. doi: 10.1038/s41576-018-0032-z. PMID: 29973680.

Why bacterial population structure matters?

- Reveals lineage-specific patterns.
- Enables transmission and outbreak detection.
- Facilitates the investigation of bacterial evolution, adaptation and diversification
- Provide a preliminary genetic group analysed for recombination, phylogenetic or molecular dating.

Approaches to inferring bacterial relatedness

Serotyping / Pathotyping

- Surface antigens(e.g *E. coli* O157:H7), disease-causing characteristics (e.g. EAEC, EPEC etc.).

Phylogrouping

- Different evolutionary groups e.g. A, B1, C, D, E, F and others.

MLST

- Compares the sequences of 7 house keeping genes. Combination alleles define ST.

Whole-genome sequence analysis

- Compares thousands of gene across the core genomes. It gives higher resolution (e.g. BAPs).

Bayesian Population Structure

- A statistical approach that uses genome-wide genetic variation to infer genetically related population without using a predefined typing scheme.

Key features

- No predefined genetic groups.
- Uses genome-wide nucleotide variation.
- Identifies hidden population structure.

Framework of Bayesian population structure

Collections of bacterial genome

Examine Genome-wide genetic variation

Compare different possible groupings

Which best explains the observed genetic variation

Select the best grouping

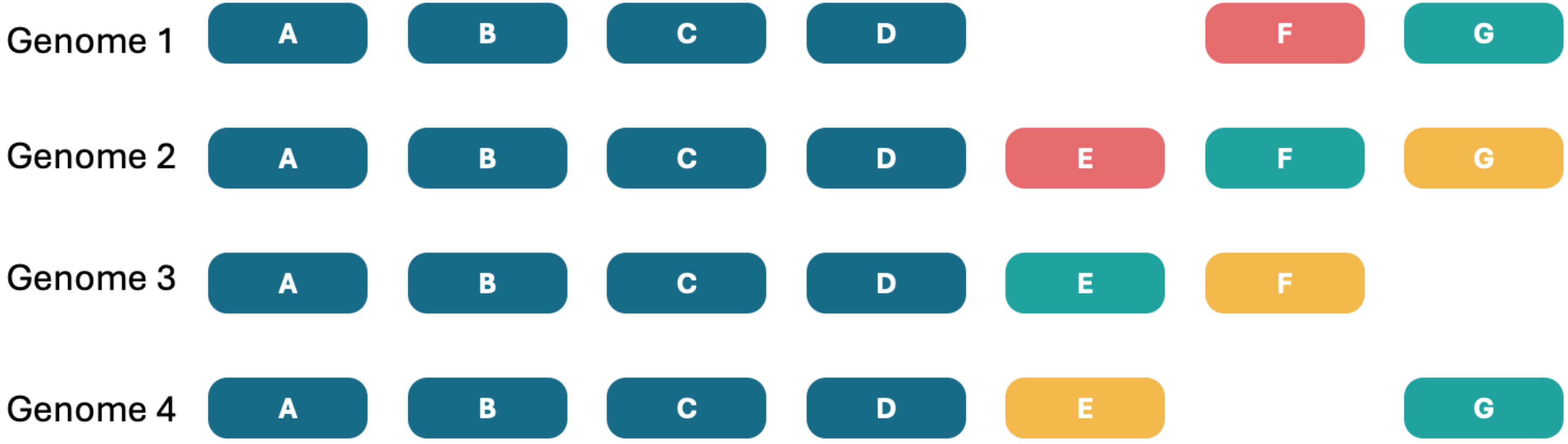
Infer population structure

Software	Description
BAPS	Original implementation of Bayesian Analysis of Population Structure (corander et al., 2003).
hierBAPS	Extends BAPS to infer hierarchical population structure (clusters within clusters) (Cheng et al., 2013).
RhierBAPS	R implementation of hierBAPS, allowing integration into R-based genomic workflows (Cheng et al., 2013).
FastBAPS	**A faster, scalable implementation designed for large bacterial genomic datasets (**Tonkin-Hil et al., 2019)**.**

The Bacterial Pangenome: Core and Accessory Genomes

Pangenome is the complete set of genes found across all genomes in a dataset.

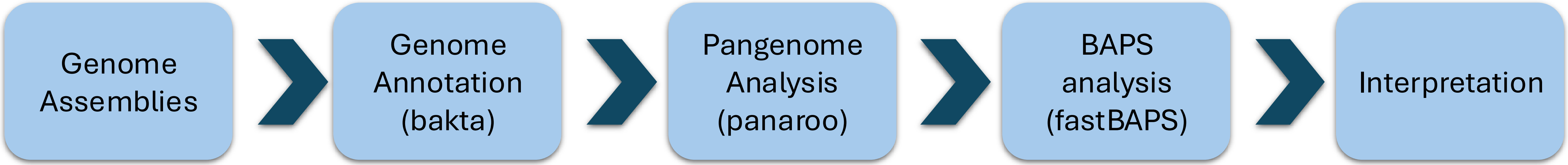
Two components : **Core genes –** all (nearly all) genomes &
Accessory genes – subset of genomes



Core vs Accessory genes

Core Genome	**Accessory Genome**
Present in all or nearly all genomes	Present in only some genomes
Largely vertically inherited	Often gained or lost through horizontal gene transfer
Shared across isolates	Variable among isolates
Suitable for genome-wide alignment	Cannot be aligned across all genomes because many genes are absent

Bayesian Population Structure workflow



References

- Sheppard SK, Guttman DS, Fitzgerald JR. Population genomics of bacterial host adaptation. Nat Rev Genet. 2018 Sep;19(9):549-565. doi: 10.1038/s41576-018-0032-z. PMID: 29973680.
- Tonkin-Hill, Gerry, John A Lees, Stephen D Bentley, Simon D W Frost, and Jukka Corander. 2019. “Fast Hierarchical Bayesian Analysis of Population Structure.” *Nucleic Acids Res.*, May. https://doi.org/10.1093/nar/gkz361.
- Cheng, Lu, Thomas R Connor, Jukka Sirén, David M Aanensen, and Jukka Corander. 2013. “Hierarchical and Spatially Explicit Clustering of DNA Sequences with BAPS Software.” *Mol. Biol. Evol.* 30 (5): 1224–28. https://doi.org/10.1093/molbev/mst028.
- Corander J, Waldmann P, Sillanpää MJ. Bayesian analysis of genetic differentiation between populations. Genetics. 2003 Jan;163(1):367-74. doi: 10.1093/genetics/163.1.367. PMID: 12586722; PMCID: PMC1462429.

Practical Session

Diarrhoeagenic *Escherichia coli* genomes